

Week 7 – Visualisation & Storytelling

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(Individual Submission)

This file contains the required structure for your Week 7 submission. Replace all placeholder text with your own work.

1 Dataset Used

Important

You are expected to use the same dataset that you used during previous weeks. However, you may switch to another dataset if you can justify the switch. The new dataset (if switched to) must adhere to the same theme as before.

Briefly describe the dataset you are visualising in the callout below (2–4 sentences). If switching datasets, complete the note below.

Dataset Description and Link

I used the same data set. (EV_Europe_BEV_Stock_2015_2024)

The data set provides comprehensive information on global electric vehicle (EV) trends, focusing on vehicle stock, sales, charging infrastructure, and powertrain types (BEV, PHEV, FCEV) from 2010 to 2024. It includes detailed country-level data across multiple indicators such as EV adoption, policy incentives, energy use, and charging point deployment. The data set enables our team to understand various aspects of our learning from different modules and have an analytical report of the comparative analysis of EV growth, particularly across European region.

Dataset Switch Note (Only if applicable)

- Why the original dataset could not be used responsibly
- Why this new dataset fits the same theme
- Source + license

1.1 Improved Visuals from Previous Weeks

(Minimum 1, Maximum 3)

Please provide reference to the visuals you choose to improve from previous submissions with traceable and clear reference.

What “improvement” of a previous plot means (click to reveal)

1.1.1 2.1 Original Plot 1 (from Week 2 or 4)

```
options(dplyr.summarise.inform = TRUE)
knitr::opts_chunk$set(fig.width = 12,
                        fig.height = 8,
                        include = TRUE,
                        echo = TRUE)

library(tidyverse)
library(readxl)
library(dplyr)

# Step 1: Load the Dataset
# Reads the first sheet ("Page 1" in Excel)

ev_data <- read_csv("data/processed/EV_Europe_BEV_Stock_2015_2024.csv")
head(ev_data, 10)
```

region_country	year	value
<chr>	<dbl>	<dbl>
Germany	2024	1900000
Germany	2023	1500000
United Kingdom	2024	1400000
France	2024	1100000
Germany	2022	1000000
France	2023	980000
United Kingdom	2023	980000
Norway	2024	750000
Norway	2023	690000
France	2022	620000

1-10 of 10 rows

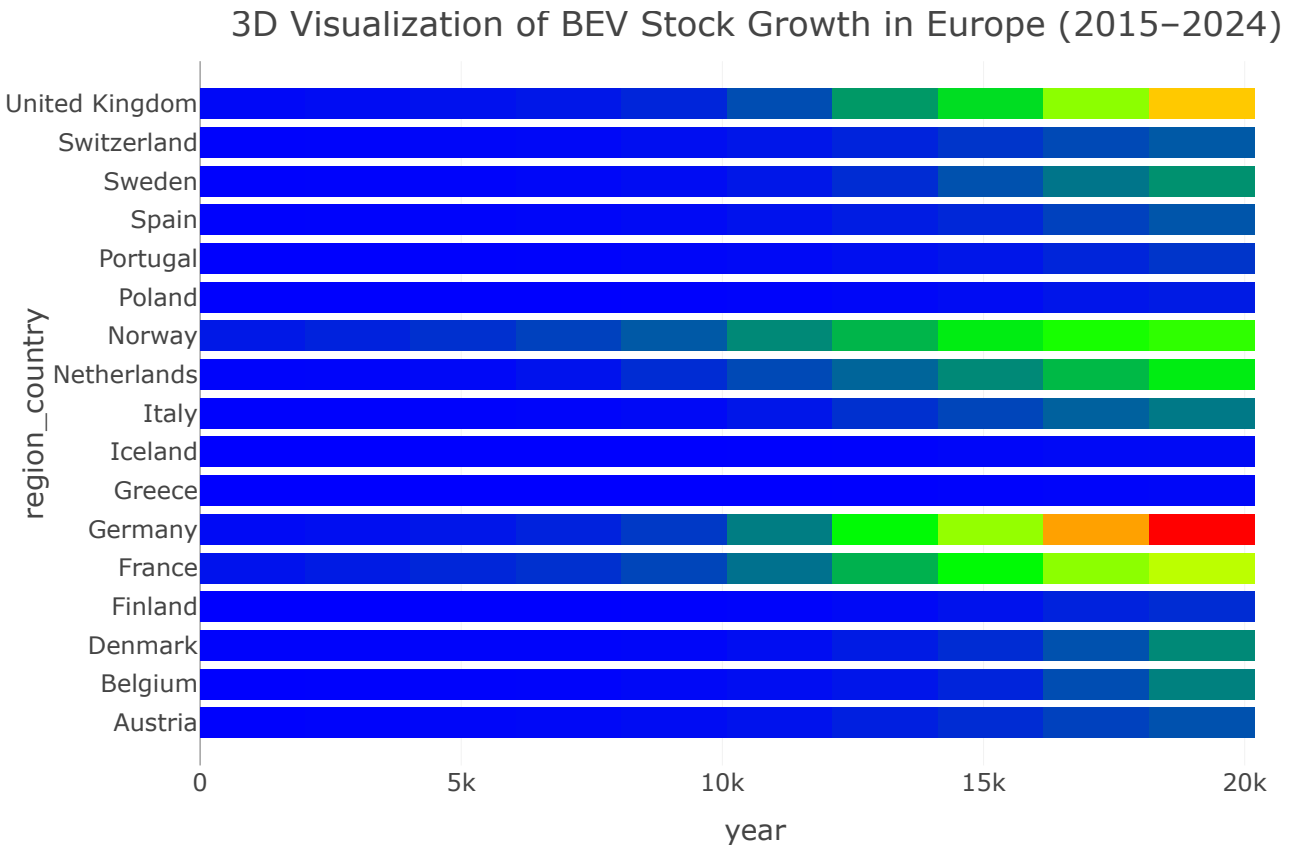
```
# Step 4.3.1 : 3D Visualization using Plotly
library(plotly)

#Aggregate data to reduce clutter (sum by year & country)
ev_agg <- ev_data %>%
  group_by(region_country, year) %>%
  summarise(Total_Stock = sum(value, na.rm = TRUE)) %>%
  ungroup()

# 3D Scatter Plot
plot_ly(
  ev_agg,
  x = ~year,                # X-axis: Time
  y = ~region_country,      # Y-axis: Country names
  z = ~Total_Stock,         # Z-axis: EV stock
  type = "bar",

  color = ~Total_Stock,    # Color by EV stock intensity
  colors = colorRamp(c("blue", "green", "yellow", "red"))
) %>%
```

```
layout(
  title = "3D Visualization of BEV Stock Growth in Europe (2015–2024)",
  scene = list(
    xaxis = list(title = "Year"),
    yaxis = list(title = "Country"),
    zaxis = list(title = "EV Stock (Units)")
  )
)
```



Include a short reminder of what the original plot attempted to show.

The plot wanted to show 3 dimensions of ev data in a single graph(Year, Countries and EV_sales). It was done by using a 3D color map, year in X-axis , countries as Y-axis and colors map for the values. The issue with it is, year has been summarized so x axis is not actually in years. Next, we cannot define the growth with this color map for each countries. We can only say that it is increasing with time and some countries better than other. And scatter plot is better visualization tool for this task.

1.1.2 2.1 Improved Plot 1 (Week 7)

```
ev_wide <- ev_data %>%
  select(region_country, year, value) %>%
  group_by(region_country, year) %>%
  summarise(value = sum(value, na.rm = TRUE)) %>%
  ungroup() %>%

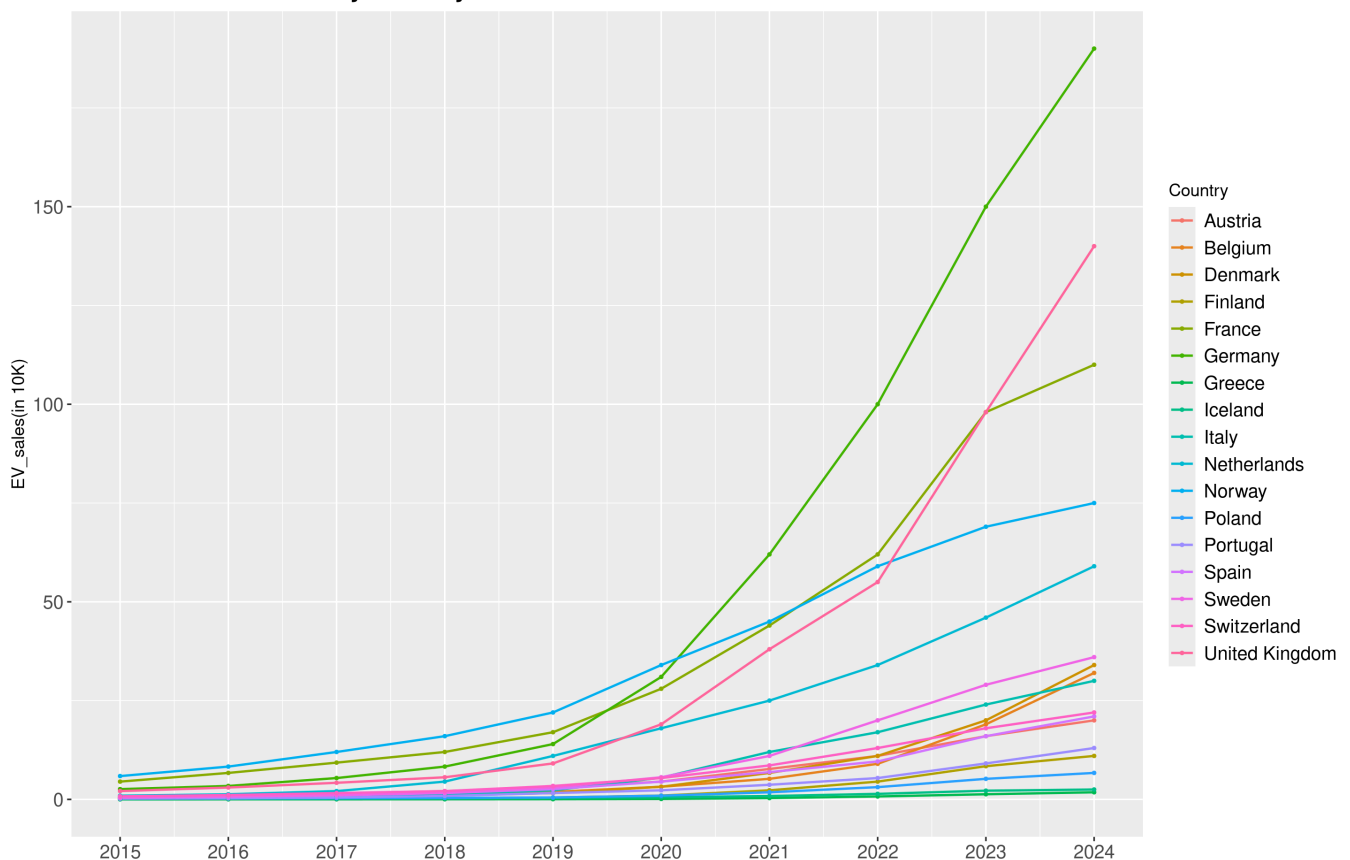
  pivot_wider(names_from = year, values_from = value)
ev_long <- ev_wide %>%
  pivot_longer(
    cols = -region_country,
```

```

    names_to = "year",
    values_to = "value"
  ) %>%
  mutate(year = as.numeric(year))

# Plot values as function of time
ggplot(ev_long, aes(x = year, y = value/10000, color = region_country, group = region_
  geom_line(size = 0.7) +
  geom_point(size = 0.7) +
  scale_x_continuous(breaks = c(min(ev_long$year):max(ev_long$year))) +
  #theme_minimal(base_size = 14) +
  labs(
    title = "EV Metrics Over Time by Country",
    y = "EV_sales(in 10K)",
    color = "Country"
  ) +
  theme(
    legend.position = "right",
    legend.text = element_text(size = 12),
    plot.title = element_text(face = "bold", size = 16),
    axis.text.y = element_text(size = 12),
    axis.text.x = element_text(size = 12),
    axis.title.x = element_blank(),
  )

```

EV Metrics Over Time by Country

```

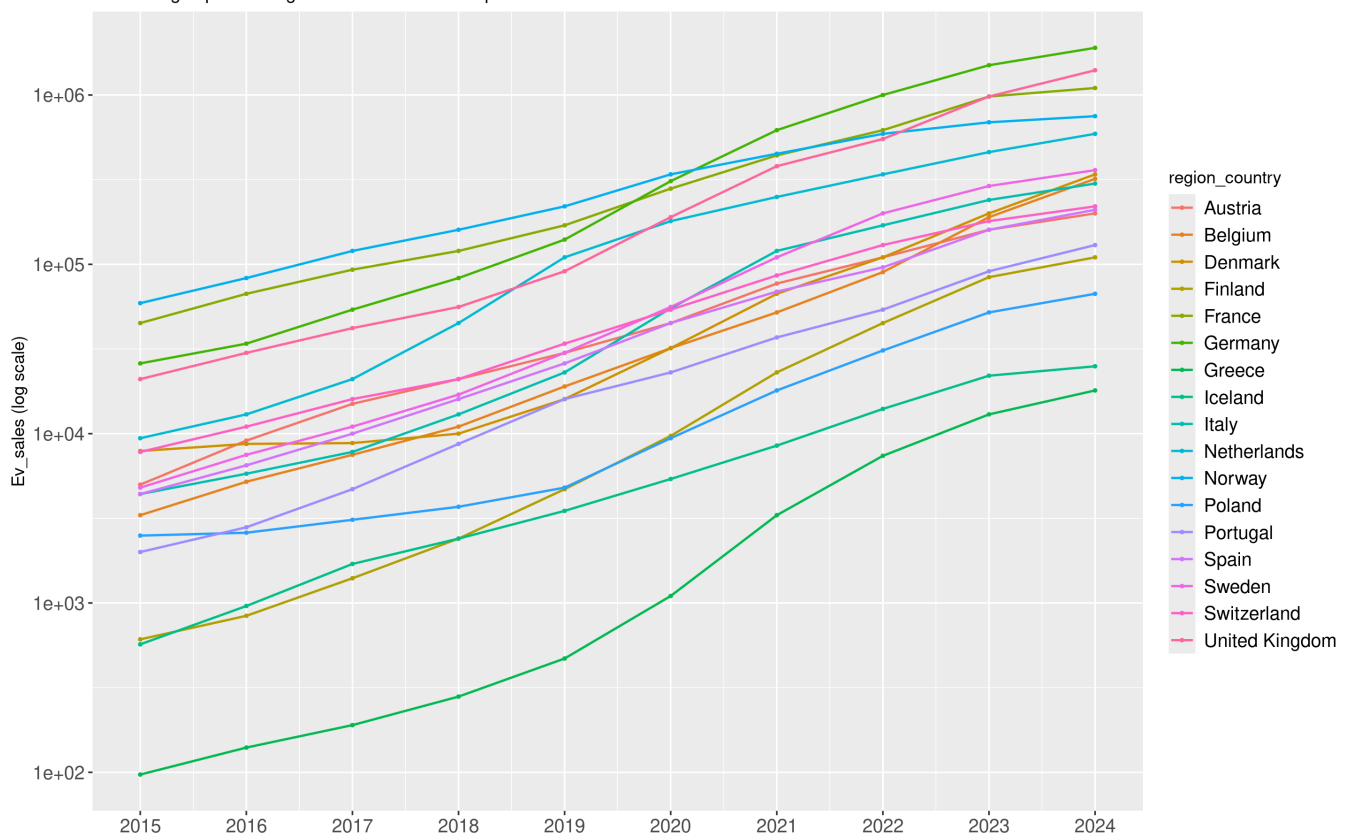
# Plot values as function of time in log scale
ggplot(ev_long, aes(x = year, y = value, color = region_country, group = region_countr
  geom_line(size = 0.7) +

```

```
geom_point(size = 0.7) +
scale_y_log10() +
scale_x_continuous(breaks = c(min(ev_long$year):max(ev_long$year))) +
#theme_minimal(base_size = 14) +
labs(
  title = "EV Growth Over Time by Country (Log Scale)",
  subtitle = "Visualizing exponential growth trends in EV adoption",
  y = "Ev_sales (log scale)",
  color = "region_country"
) +
theme(
  legend.position = "right",
  legend.text = element_text(size = 12),
  plot.title = element_text(face = "bold", size = 16),
  axis.text.y = element_text(size = 12),
  axis.text.x = element_text(size = 12),
  plot.subtitle = element_text(size = 12),
  axis.title.x = element_blank(),
)
```

EV Growth Over Time by Country (Log Scale)

Visualizing exponential growth trends in EV adoption



What I Improved and Why (3–6 bullet points):

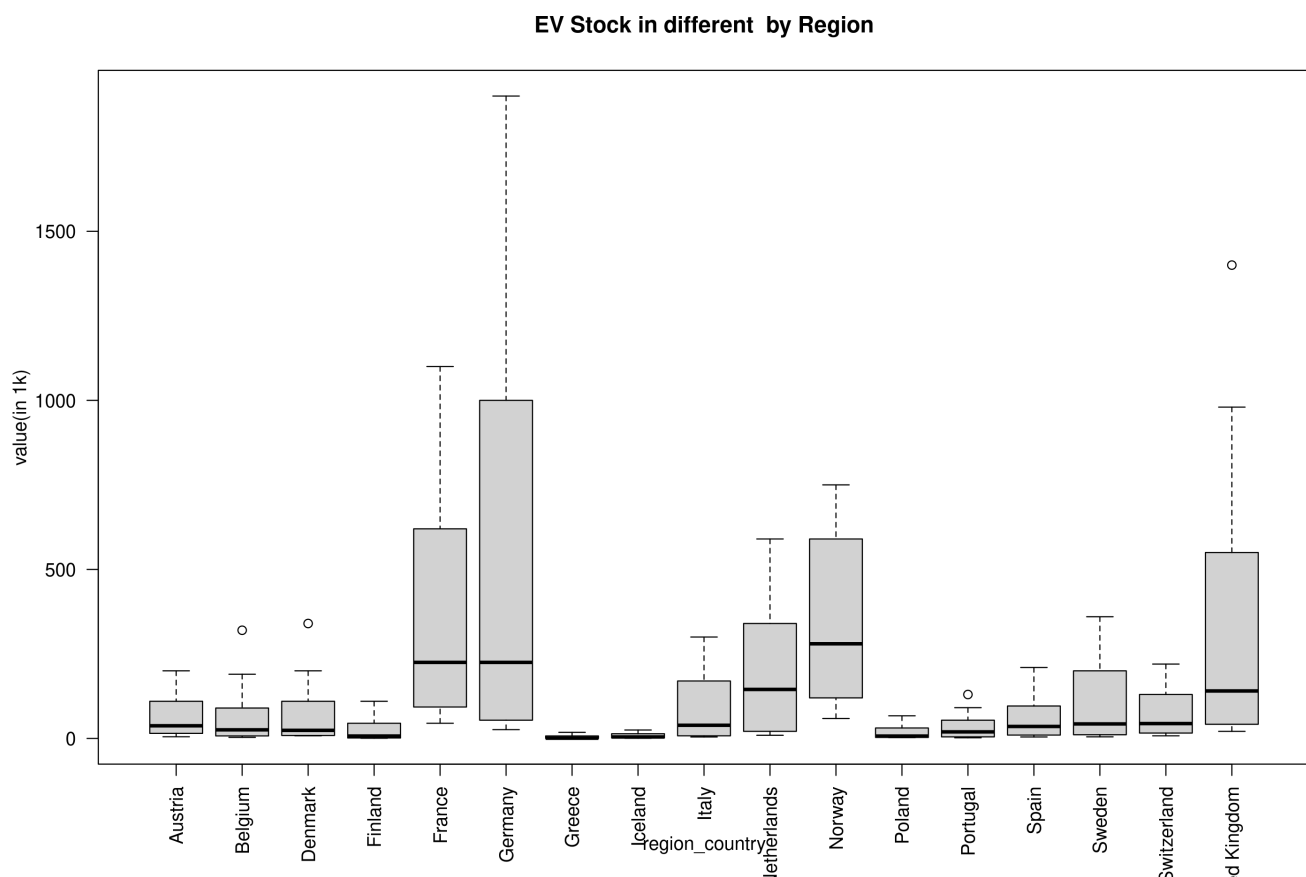
- The scatter plot shows clear difference in the growth curve for each countries.
- From the previous graph in week 2, adjust the line,dot width and added better theme to make it more visually understanding.
- From the previous graph in week 2, Y labels were changed from “values” to “ev sales in 10K”

- The scatter plot with log scale is also added, the bottom region of the graph is very cluttered in normal scale and to have better understanding for the bottom region, log scale visualization give a better understanding. (for eg, Greece even though had slow start but had tremendous growth from 2015 to 2024)

Include require text/code sections if you are adding more improved visuals from previous weeks.

1.1.3 2.2 Original Plot 2 (from Week 2 or 4)

```
boxplot(
  value/1000 ~ region_country,
  data = ev_long,
  main = "EV Stock in different by Region",
  xlab = "region_country", ylab = "value(in 1k)",
  las = 2 # rotate x labels if long
)
```



Include a short reminder of what the original plot attempted to show.

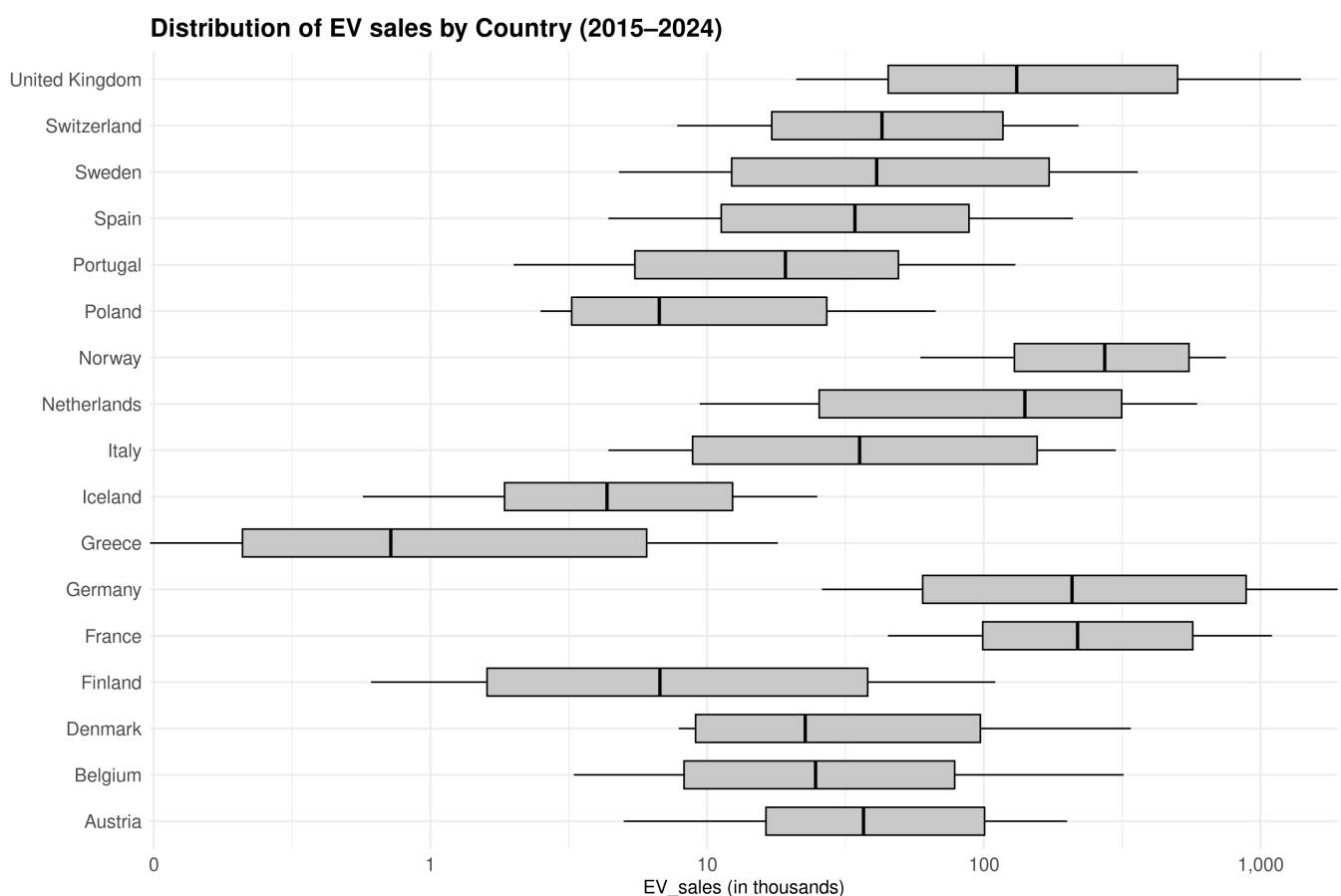
The box-plot shows the ev-sales as minimum, lower quartile (Q1), median (Q2), upper quartile (Q3), and maximum and outliers for different region. The problem is x labels are cluttered and overlapping. The regions with comparatively less sales cannot be easily understood (eg, Greece, Iceland, Finland, etc) in normal scale as their values is very less compared to top 5.

1.1.4 2.2 Improved Plot 2 (Week 7)

```
library(ggplot2)
library(scales)

p_box <- ggplot(ev_data, aes(x = region_country, y = value/1000)) +
  geom_boxplot(fill = "grey80", color = "black", width = 0.6) +
  coord_flip() +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.text.y = element_text(size = 12),
    axis.text.x = element_text(size = 12),
    axis.title.y = element_blank(),
  ) +
  labs(
    title = "Distribution of EV sales by Country (2015–2024)",
    x = "",
    y = "EV_sales (in thousands)"
  ) +
  scale_y_log10(labels = comma_format(accuracy = 1), expand = c(0,0))

print(p_box)
```



What I Improved and Why (3–6 bullet points):

- The improved box plot with log scales shows better visual representation for all countries.
- The coordinators were flipped to give more space for each region and is now less cluttered
- The box plot with log scale for sales is also added, the countries with less sales cannot be easily understood (eg, Greece, Iceland, Finland, etc) in normal scale and to have better understanding for the these region, log scale

visualization give a better understanding. (for eg, Greece even though had had tremendous growth from 2015 to 2024)

1.2 New Week 7 Visuals

You must produce **at least two new required visuals**:

Possible New Plot Types for Week 7 (Click to reveal)

Note: If required plots are not possible, please justify the alternatives and adjust the text and explanations accordingly.

1.2.1 A. Relationship Visual (Required)

Examples: scatterplot + trend line, scatter by subgroup, faceted scatter.

```
ev_data <- ev_data %>%
  mutate(
    year = as.numeric(year),
    region_country = as.character(region_country)
  )

# defining regions

west_countries <- c("Germany", "France", "United Kingdom", "Netherlands",
  "Belgium", "Austria", "Switzerland", "Ireland")
nordic_countries <- c("Sweden", "Finland", "Denmark", "Norway", "Iceland")
south_countries <- c("Italy", "Spain", "Portugal")
east_countries <- c("Poland", "Czech Republic", "Slovakia", "Hungary",
  "Romania", "Bulgaria", "Greece")

ev_data <- ev_data %>%
  mutate(
    region = case_when(
      region_country %in% west_countries ~ "West",
      region_country %in% nordic_countries ~ "Nordic",
      region_country %in% east_countries ~ "East",
      region_country %in% south_countries ~ "South",
      TRUE ~ NA_character_
    )
  )

# facets order
ev_data$region <- factor(ev_data$region, levels = c("West","Nordic","East","South"))

# group by top 3 countries
top3_by_region <- ev_data |>
  group_by(region, region_country) |>
```

```

summarise(total_value = sum(value), .groups = "drop") |>
arrange(region, desc(total_value)) |>
group_by(region) |>
slice_head(n = 3) |>
ungroup()

top3_list <- top3_by_region$region_country

ev_top3 <- ev_data |>
  filter(region_country %in% top3_list)

# loess smoothing by region

regional_loess_data <- ev_data |>
  group_by(region, year) |>
  summarise(region_total = sum(value))

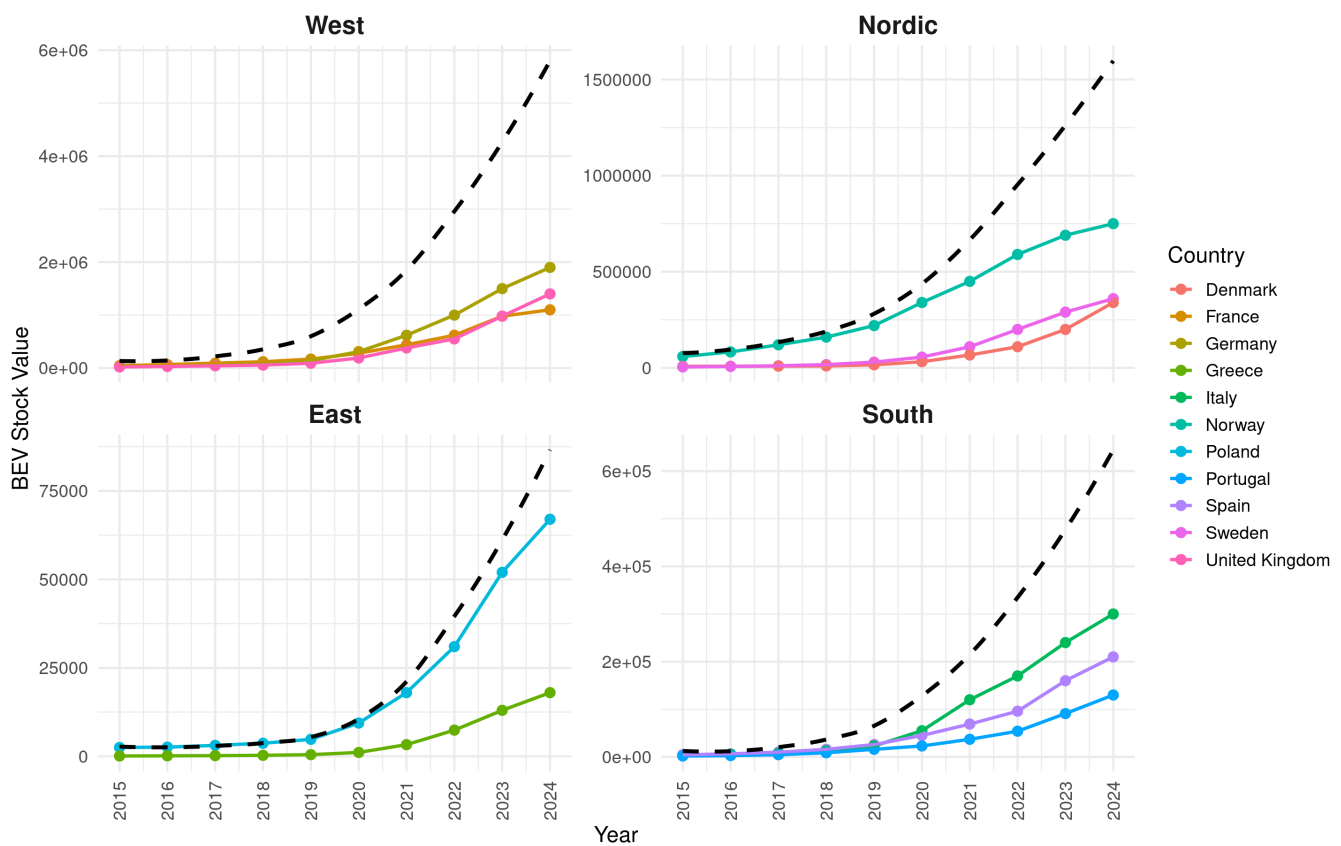
# 5. Plot

ggplot() +
  geom_point(
    data = ev_top3,
    aes(x = year, y = value, color = region_country),
    size = 2.7
  ) +
  geom_line(
    data = ev_top3,
    aes(x = year, y = value, color = region_country, group = region_country),
    linewidth = 1
  ) +
  geom_smooth(
    data = regional_loess_data,
    aes(x = year, y = region_total, group = region),
    method = "loess",
    se = FALSE,
    linewidth = 1.2,
    color = "black",
    linetype = "dashed",
    alpha = 0.8
  ) +
  facet_wrap(~ region, scales = "free_y") +
  theme_minimal(base_size = 14) +
  labs(
    title = "Top 3 Countries per Region with reigon Level LOESS Trend",
    x = "Year",
    y = "BEV Stock Value",
    color = "Country"
  ) +
  scale_x_continuous(
    breaks = seq(2015, 2024, 1),
    labels = seq(2015, 2024, 1)
  ) +
  theme(

```

```
strip.text = element_text(size = 16, face = "bold"),
axis.text.x = element_text(angle = 90, vjust = 0.5)
)
```

Top 3 Countries per Region with region Level LOESS Trend



1.2.2 If you used AI:

Include the following block for each interaction.

- Tool/model used: Chapgpt
- Prompt (short paraphrase is OK): Mostly to understand the code to derive faceted graphs for each region and also for losses estimates
- What the AI generated: Generated a code fro facets
- How you verified it: by checking with the actual data
- What you changed: implemented in my code using my data set

Interpretation (2–4 sentences):

The graph shows the LOESS-smoothed curves for top 3 countries in each region from 2015 to 2024. It shows that that Western Europe has by far the largest and fastest-growing BEV stock, accelerating sharply after 2020. The Nordic region also grows strongly, but at a lower absolute scale.. The LOESS-smoothed curves show that Western Europe has by far the largest and fastest-growing BEV stock, accelerating sharply after 2020. The South and East regions have slower growth, starting from much smaller base levels. Overall, the plot reveals distinct regional trajectories and widening gaps between regions over time.

Why this plot type is appropriate (1–2 sentences): Individual countries vary widely. LOESS smooths the data and aggregates by region, making long-term patterns clearer. It highlights structural differences in EV adoption between differnt regions. It summarize complex data on one clean chart.

1.2.3 B. Distribution Comparison (Required)

Choose one: boxplot, violin, density comparison, faceting.

```
# total EV stock by country
ev_data <- ev_data |>
  mutate(year = factor(year))

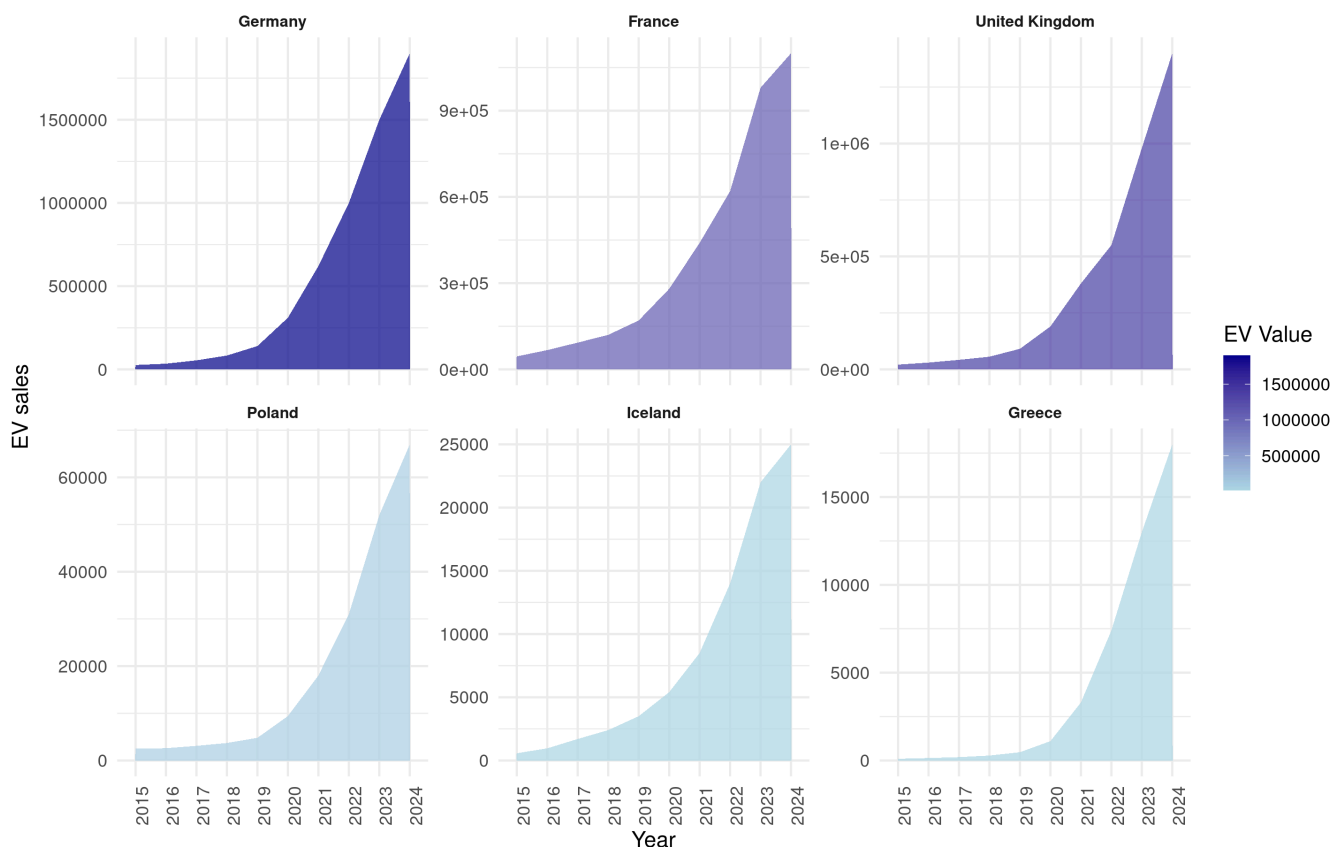
totals <- ev_data |>
  group_by(region_country) |>
  summarise(total_stock = sum(value)) |>
  arrange(desc(total_stock))

# top 3 and bottom 3 countries
top3 <- head(totals$region_country, 3)
bottom3 <- tail(totals$region_country, 3)

# facet order: top 3 first, bottom 3 last
facet_order <- c(top3, bottom3)

ev_sub <- ev_data |>
  filter(region_country %in% facet_order) |>
  mutate(region_country = factor(region_country, levels = facet_order))

# plot
ggplot(ev_sub, aes(
  x = year,
  y = value,
  fill = value,
  group = region_country
)) +
  geom_area(alpha = 0.7) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  facet_wrap(~ region_country, scales = "free_y", nrow = 2) +
  theme_minimal(base_size = 14) +
  labs(
    title = "BEV Sales Distribution (Top 3 & Bottom 3 Countries)",
    x = "Year",
    y = "EV sales",
    fill = "EV Value"
  ) +
  theme(
    strip.text = element_text(size = 10, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 90, hjust = 1)
  )
```

BEV Sales Distribution (Top 3 & Bottom 3 Countries)**1.2.4 If you used AI:**

Include the following block for each interaction.

- Tool/model used: Chapgpt
- Prompt (short paraphrase is OK): Mostly to understand the code to derive faceted graphs for each group
- What the AI generated: Generated a code for facets
- How you verified it: by checking with the actual data
- What you changed: implemented in my code using my data set

Interpretation (2–4 sentences):

Germany, France and the UK, the top, show extremely rapid BEV sales especially after 2020, reflecting strong EV adoption and supportive policies.

In contrast, Poland, Iceland, and Greece, the bottom three, exhibit much smaller BEV sales and slower growth. The gap between the two groups widens over time, indicating increasing divergence in EV market maturity across Europe.

Why this plot type is appropriate (1–2 sentences):

Instead of overlapping lines or stacked areas, each country gets its own smooth density ridge, reducing visual noise while keeping patterns comparable.

Top countries (Germany, France, the UK) have values in the millions, while the bottom ones (Iceland, Greece, Poland) are much smaller.

Ridge plots preserve the form of the growth curve without the smaller countries being visually crushed.

1.2.5 C. Trend / Group Comparison (Optional)

```

ev_data <- ev_data %>%
  mutate(
    year = as.numeric(year),
    region_country = as.character(region_country)
  )

# defining regions

west_countries <- c("Germany", "France", "United Kingdom", "Netherlands",
  "Belgium", "Austria", "Switzerland", "Ireland")
nordic_countries <- c("Sweden", "Finland", "Denmark", "Norway", "Iceland")
south_countries <- c("Italy", "Spain", "Portugal")
east_countries <- c("Poland", "Czech Republic", "Slovakia", "Hungary",
  "Romania", "Bulgaria", "Greece")

ev_data <- ev_data %>%
  mutate(
    region = case_when(
      region_country %in% west_countries ~ "West",
      region_country %in% nordic_countries ~ "Nordic",
      region_country %in% east_countries ~ "East",
      region_country %in% south_countries ~ "South",
      TRUE ~ NA_character_
    )
  )

# facets order
ev_data$region <- factor(ev_data$region, levels = c("West","Nordic","East","South"))

# group by top 3 countries
top3_by_region <- ev_data |>
  group_by(region, region_country) |>
  summarise(total_value = sum(value), .groups = "drop") |>
  arrange(region, desc(total_value)) |>
  group_by(region) |>
  slice_head(n = 3) |>
  ungroup()

top3_list <- top3_by_region$region_country

ev_top3 <- ev_data |>
  filter(region_country %in% top3_list)

# Summary of LOESS fits per region
regional_loess_data <- ev_data |>
  group_by(region, year) |>
  summarise(region_total = sum(value))

loess_summary <- regional_loess_data |>
  group_by(region) |>
  group_modify(~{
    fit <- loess(region_total ~ year, data = .x)
  })

```

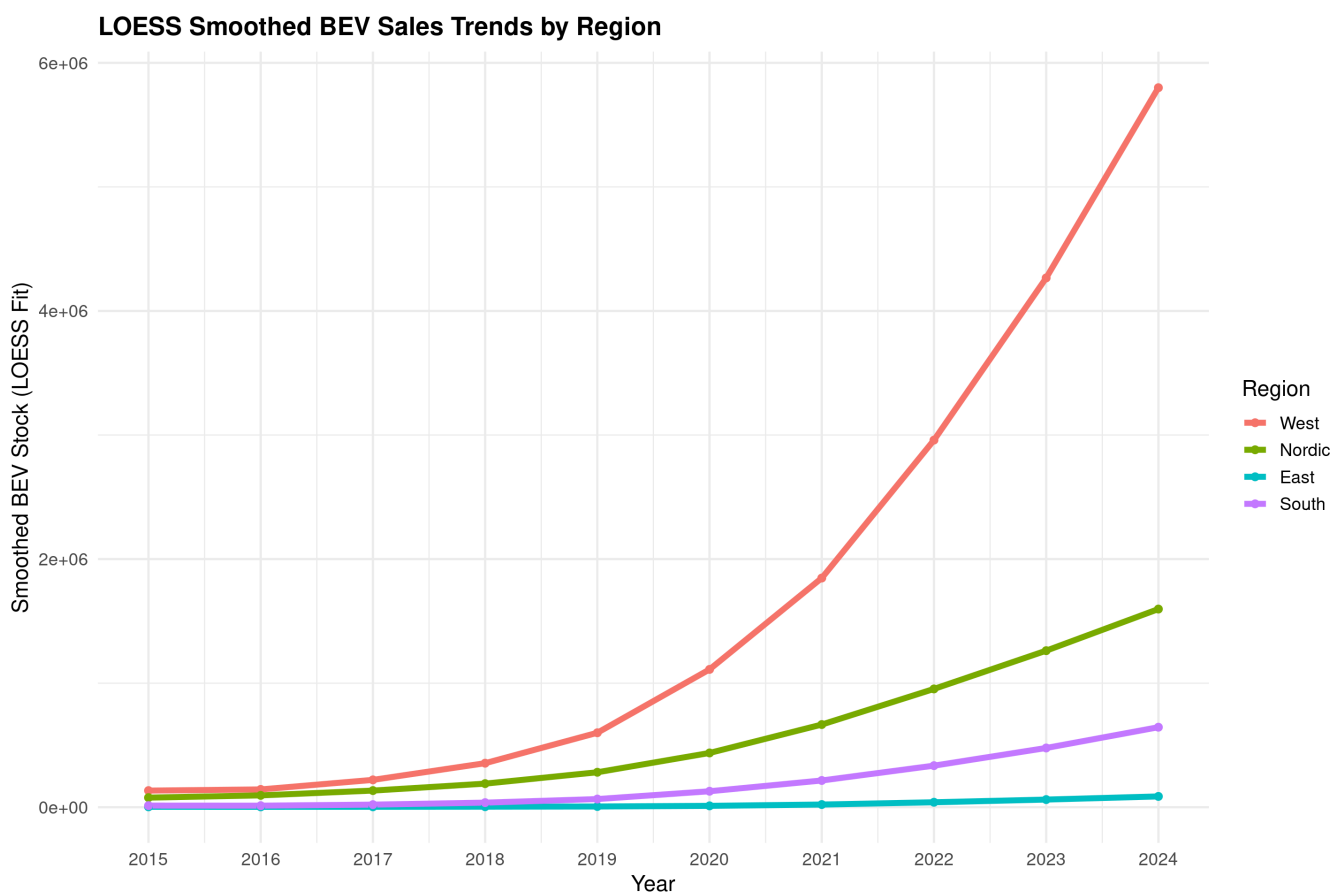
```

  tibble(
    year = .x$year,
    loess_fit = predict(fit, .x$year)
  )
})

ggplot(loess_summary, aes(x = year+2014, y = loess_fit, color = region)) +
  geom_line(linewidth = 1.8) +
  geom_point(size = 2) +
  theme_minimal(base_size = 14) +
  labs(
    title = "LOESS Smoothed BEV Sales Trends by Region",
    x = "Year",
    y = "Smoothed BEV Stock (LOESS Fit)",
    color = "Region",
  ) +
  scale_x_continuous(
    breaks = seq(2015, 2024, 1),
    labels = seq(2015, 2024, 1)
  ) +

  theme(
    legend.position = "right",
    plot.title = element_text(size = 16, face = "bold")
  )

```



1.2.6 If you used AI:

Include the following block for each interaction.

- Tool/model used: Chapgpt
- Prompt (short paraphrase is OK): Mostly to understand the code for losses estimates
- What the AI generated: Generated a code fro facets
- How you verified it: by checking with the actual data
- What you changed: implemented in my code using my data set

Interpretation (2–4 sentences):

The graph shows the LOESS-smoothed curves for top 3 countries in each region from 2015 to 2024 as groups. Overall, the plot reveals distinct regional trajectories and widening gaps between regions over time.

Why this plot type is appropriate (1–2 sentences): It highlights structural differences in EV adoption between differnt regions. It summarize complex data on one clean chart.

1.3 Narrative [🔗](#)

Write a short narrative connecting your visuals into a coherent data story.

Example structure (keep it simple):

- What question/theme the visuals address
- What the relationship plot reveals
- What the distribution plot clarifies
- Optional: what trend/group plot adds
- A closing insight

This set of visuals explores how Battery Electric Vehicle (BEV) sales have evolved across European regions from 2015 to 2024, with a particular focus on identifying which countries are driving growth within each region. By narrowing the view to the top three countries in every region, the analysis highlights both regional patterns and national contributions.

The relationship plot shows country level BEV growth within each geographic group. It quickly becomes clear that Western and Nordic countries exhibit strong, accelerating adoption, while Eastern and Southern regions grow more steadily but from smaller baselines.

The addition of region level LOESS curves clarifies the broader trajectories behind individual country lines. These smoothed trends show that overall growth is steepest in the West, followed closely by the Nordic region, whereas the East and South display later and more gradual adoption waves. The contrast between individual country variability and the smooth regional curve helps separate national outliers from underlying regional momentum.

Together, these visuals reveal a clear pattern, Europe is converging toward higher BEV adoption, but at different speeds. A small set of high adoption countries especially in the West and Nordic regions—are responsible for pulling regional trends upward, while other areas are still in early growth phases. This suggests that policy maturity, charging infrastructure, and economic capacity remain key drivers of regional divergence in EV transition.

1.4 Responsible Design Notes

Provide brief bullet points about your design decisions:

PLot 1 & 2 * Logarithmic y-axis: I used a log scale to clearly reveal exponential-style growth, prevent larger countries from visually overpowering smaller ones, and allow all countries to be compared on a consistent rate-of-change basis.

- Thin lines + high color contrast: Because there are many countries, lightweight line styling and a diverse but balanced color palette reduce visual clutter while still allowing each country to remain distinguishable.
- Minimal grid + clear titles: Light gridlines and a bold, descriptive title provide orientation without overwhelming the data, helping viewers immediately understand that the focus is on long-term growth trends rather than individual year-to-year fluctuations.
- Used a horizontal boxplot to improve readability because countries have long names and many categories; horizontal layout prevents label overlap.
- Applied a log-scale on the x-axis to handle the extremely wide range of EV sales (from <1k to >1M), preventing large countries from dominating the visual.
- Kept the styling minimal (grayscale boxes) to avoid unnecessary color load for a plot where comparison of distributions—not categories—is the goal.
- Ordered countries by median sales to guide the viewer's eye from lowest to highest EV adoption levels and make patterns easier to interpret.
- Included clear axis labels and title to ensure viewers understand the time period and variable without needing external context.
- Filtered to the top three countries per region to avoid visual clutter and ensure fair comparison across unequal group sizes.

Plot 3 4 & 5 design choice

- Used region-level LOESS smoothing to highlight underlying long-term trends without overstating short-term fluctuations.
- Separated regions into facets to maintain readability and prevent dominant countries (e.g., West/Nordic leaders) from overwhelming smaller-scale regions.
- Applied consistent color palettes and clear labels to support accessibility and reduce cognitive load when comparing countries and regions.

1.5 Responsible Use of AI (RVE Note)

Complete one of the two options below.

1.5.1 If you used AI:

Include the following block for each interaction.

- Tool/model used: Chapgpt
- Prompt (short paraphrase is OK):
- What the AI generated:
- How you verified it:
- What you changed:

1.5.2 If you did NOT use AI:

"I did not use AI for code or interpretation. All design decisions and commentary are my own."

1.6 Appendix (Optional)

You may place additional plots, tests, or reproducibility info here.

```
# Optional extras
sink("session-info.txt") # redirect console output to a file
cat("=== Reproducibility Log ===\n")
cat("Date:", Sys.Date(), "\n\n")
sessionInfo()           # records R version, OS, and packages
sink()                  # stop redirection

# Check that the log file was created
file.exists("session-info.txt")
```

[1] TRUE

2 END OF FILE